



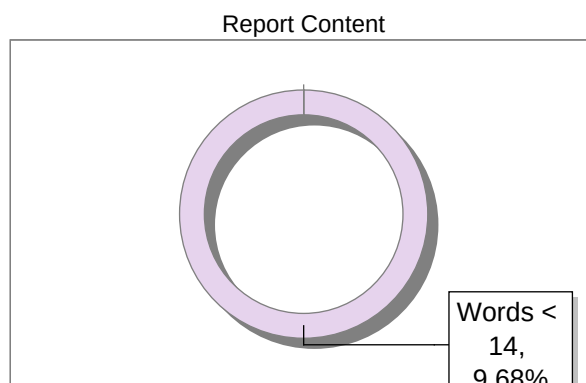
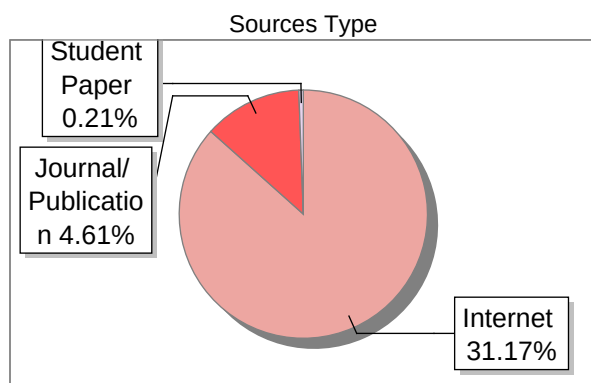
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IV. EXPERIMENTAL RESULTS

In this experiment, we have evaluated 102 cowpea germplasm lines along with five checks viz., C-152, KBC-2, KBC-9, IT-38956-11 and a local variety for diversity, variability, and association among the traits through morphological characterization for seed yield and its attributing traits. Further, the cowpea germplasm lines were characterized based on seed morphometric and quality traits during *Kharif-2025*.

The following section represents the results obtained from the present study under the following headings:

- 4.1 To study the genetic variability for grain yield and its attributing quantitative traits.
- 4.2 To study the genetic divergence and associationship between grain yield and its attributing quantitative traits.
- 4.3 To study the seed morphometric and quality traits of cowpea genotypes.
- 4.1 To study the genetic variability for grain yield and its attributing quantitative traits.

The study evaluated 102 cowpea germplasm lines, along with five check varieties, during *Kharif-2025* using an Augmented Randomized Block design with six blocks to determine the variability in seed yield and yield attributing traits among cowpea germplasm lines. Findings of the above mentioned objectives are presented below with subheadings mentioned as:

- 4.1.1. Analysis of variance.
- 4.1.2. Genetic variability analysis for seed yield and its attributing traits.
- 4.1.3. Principal component analysis for seed yield and its attributing traits.

4.1.1 Analysis of variance (ANOVA)

Analysis of variance (ANOVA) is a statistical technique used to determine whether significant differences exist among the means of different groups. Analysis of variance of 12 quantitative characters was performed among 107 cowpea genotypes and ANNOVA is in Table 3 and Table 4. Analysis displayed significant differences among the cowpea germplasm lines for all quantitative characters studied, indicating the existence of sufficient variability among the cowpea genotypes.

ANNOVA studies revealed that the genotypes were significantly different for all the traits studied. Similarly, the effects due to treatment, due to check and due to check and variety were also significant for most of the traits. This shows that there is a significantly higher amount of variability among the cowpea genotypes. Similar findings were also observed by Poornima *et al.* (2023), Chipeta *et al.* (2024) and Niji *et al.* (2025).

4.1.2. Genetic variability analysis for seed yield and its attributing traits.

Genetic variability is the foundation of any crop improvement programme, as it provides the opportunity to identify and select superior genotypes with desirable agronomic traits. Therefore, the assessment of genetic parameters such as genotypic coefficient of variation (GCV), phenotypic coefficient of variation (PCV), heritability and genetic advance is essential for understanding the nature of gene action and the effectiveness of selection (Johanson *et al.* 1955).

Broad-sense heritability with high genetic advance generally indicates the predominance of additive gene action, suggesting that direct selection for such traits would be effective in developing improved cowpea genotypes.

In this regard, the present study attempted to estimate the variability components, such as mean, range for all the yield contributing traits and these are presented in Table 5 for 107 cowpea germplasm lines. The estimates of genotypic coefficient of variation (GCV), phenotypic coefficient of variation (PCV), heritability (h^2) and genetic advance as per cent of mean (GAM) for all the yield contributing traits are presented in Table 5 (Fig.1, Fig.2 and Fig.3) for 107 cowpea germplasm lines. Higher phenotypic coefficient of variation

72 values than genotypic coefficient of variation values indicated environmental influence on all traits. However, narrow differences between PCV and GCV values are observed across all traits.

47 4.1.2.1. Days to 50 per cent flowering

The days to 50 per cent flowering among the 107 cowpea genotypes ranged from 38 to 62 days, with a mean of 44 days, indicating the presence of variability in flowering time among the evaluated genotypes. The trait exhibited moderate values for PCV (11.45%) and GCV (11.24%) a narrow difference between the two estimates, suggested 3 that environmental influence on the expression of flowering time was less. High broad-sense heritability (96.47%) and a high GAM (22.78%) suggested 27 phenotypic variation was genetically controlled.

Similar findings were obtained by Panchta *et al.* (2020), Thapa *et al.* (2021), Owusu *et al.* (2022), Poornima *et al.* (2023), Abha *et al.* (2024) and Swathi *et al.* (2024). The moderate GCV and PCV, along with high heritability and high GAM, indicate 51 the presence of moderate variability with predominance of additive gene action. Therefore, direct selection would be effective for this trait.

4.1.2.2. Days to physiological maturity

The trait days to physiological maturity ranged from 80 days to 108 days, with a mean of 90 days. The trait showed lower PCV (5.21%) and GCV (5.16%) with a very narrow difference between them, indicating limited environmental influence and relatively low genetic variability for this trait. The high broad-sense heritability (97.96%) indicated that most of the observed variation was genetically determined and moderate GAM (10.5%) indicating limited variability attributable to 23 both additive and non-additive gene effects.

Therefore, selection for maturity duration can be practiced, but considerable improvement require the use of genetically diverse parents. The present findings are consistent with those reported by Owusu *et al.* (2022), Poornima *et al.* (2023) and Swathi *et al.* (2024).

4.1.2.3. Plant height (cm)

The plant height ranged from 31.46 cm to 94.31 cm, with a mean of 64.5 cm and exhibited high GCV (21.08%) and PCV (21.12%) and high broad-sense heritability (99.64%) with high genetic advance as per cent of mean (43.41%) indicated substantial heritable variation and a high expected response to selection.

The narrow difference between GCV and PCV indicates minimal environmental influence on the expression of the trait. High heritability coupled with high GAM indicates the predominance of additive gene action. Similar findings were reported by Poornima *et al.* (2023), Abha *et al.* (2024), Deeksha *et al.* (2024) and Niji *et al.* (2025).

4.1.2.4. Number of branches per plant

For the trait, number of branches per plant among 107 cowpea genotypes observed a range of 3 to 14 branches per plant, with a mean of 8 branches per plant. The trait exhibited high GCV (27.19%) and PCV (27.33%) along with high broad-sense heritability (99.03%) and GAM (55.83%) indicating the presence of high variability with additive gene action, making it suitable to improve through selection.

With this we can conclude that the environmental effect was less and a large proportion of heritable variation is present for number of branches per plant. These observations are in accordance with the findings of Panchta *et al.* (2020), Thapa *et al.* (2021), Yasin *et al.* (2021) and Poornima *et al.* (2023).

4.1.2.5. Seed length (mm)

The present study showed that the trait seed length ranged from 4.84 mm to 9.74 mm, with mean of 6.80 mm. Moderate GCV (10.03%) and PCV (10.97%) was recorded for this trait suggesting that genetic factors contributed substantially to the observed variation, although some environmental influence was present and shown high broad-sense heritability

(83.2%) with moderate GAM (18.93%) which suggested the involvement of both additive and non-additive gene action, indicating moderate scope for improvement through selection.

Thus, seed length appears to possess sufficient heritable variation for improvement, although the moderate genetic advance indicates that selection produce a moderate rather than large change in the trait. Comparable findings were reported by Porbeni *et al.* (2017), Poornima *et al.* (2023) and Joshi *et al.* (2024).

4.1.2.6. Seed breadth (mm)

The seed breadth among the 107 cowpea genotypes ranged from 3.86 mm to 7.25 mm, with a mean of 5.04 mm. The low GCV (9.22%) and moderate PCV (10.78%) the relatively larger difference between the two indicated greater environmental influence on seed breadth. High broad-sense heritability (73.03%) coupled with moderate GAM (16.25%) suggested the involvement of both additive and non-additive gene action, indicating moderate response to selection.

Similar findings were obtained by Porbeni *et al.* (2017), Yasin *et al.* (2021) Abha *et al.* (2024) and Deeksha *et al.* (2024). Thus, seed breadth was highly heritable, the relatively lower genetic variability and greater environmental influence that restrict the magnitude of improvement achievable through direct selection.

4.1.2.7. Seed thickness (mm)

Seed thickness ranged from 3.51 mm to 8.10 mm, with mean of 4.06 mm. The moderate GCV (10.94%) and PCV (12.16%) narrow difference between them shows moderate environmental effect. High heritability (80.88%) and high GAM (20.29%) indicate adequate genetic variability governed predominantly by additive gene action, suggesting good prospects for selection.

Similar findings were reported by Porbeni *et al.* (2017), Thapa *et al.* (2021) and Owusu *et al.* (2022). The combination of high heritability and high GAM suggests that selection for seed thickness would be reasonably effective in the present material.

4.1.2.8. Seed weight (g)

The present investigation showed that seed weight ranged from 7.42g to 22.17g, with mean of 9.75g. Moderate GCV (17.88%) and PCV (18.94%) with a small difference between the estimates, indicate a relatively low influence of environmental factors on the expression of the trait. High heritability (89.16%) and high GAM (34.84%) indicate sufficient genetic variability and additive gene action, making this trait responsive to direct selection. Similar findings were reported by Poornima *et al.* (2023), Abha *et al.* (2024), Deeksha *et al.* (2024) and Niji *et al.* (2025).

Variation in seed weight contribute to differences in overall seed yield. Therefore, seed weight appears to be an important trait for selection, and genotypes possessing higher seed weight can be effectively identified based on phenotypic performance.

4.1.2.9. Pod length (cm)

The pod length among the 107 cowpea genotypes ranged from 10.36 cm to 19.68 cm, with a mean of 14.44 cm. The moderate magnitude of GCV (10.32%) and PCV (10.77%) along with high heritability (91.84%) and high GAM (20.40%) indicates that additive gene effects play an important role in the inheritance of this trait. This also shows that there is less influence of environment on the trait and would responds effectively to direct selection. Similar findings were reported by Vavilapalli *et al.* (2013), Poornima *et al.* (2023) and Niji *et al.* (2025).

Since pod length is closely associated with the number and size of seeds accommodated within the pod, genotypes with desirable pod length are useful as parents for improving yield related traits.

4.1.2.10. Number of pods per plant

The number of pods per plant varied from 11 to 18 pods per plant, with a mean of 13 pods per plant. The trait exhibited moderate GCV (10.78%) and PCV (11.40%) with high heritability (89.37%) and high GAM (21.02%) suggesting adequate genetic variability and

favourable response to selection. Similar findings were reported by Kavyashree *et al.* (2023), Abha *et al.* (2024), Deeksha *et al.* (2024) and Niji *et al.* (2025).

Smaller difference between GCV and PCV show that there was limited environmental influence on its phenotypic expression. Since the number of pods per plant is an important component of seed yield, the presence of high heritability and genetic advance makes this trait useful as a selection criterion for yield improvement.

4.1.2.11. Number of seeds per pod

The evaluated cowpea genotypes produced 12 to 25 seeds per pod, with an average of 15 seeds per pod. Moderate GCV (11.41%) and PCV (11.86%) suggested that the environment effect was very limited, high heritability (92.55%) and high GAM (22.64%) indicate that this trait is largely governed by additive gene action and can be improved effectively through direct selection. Similar findings were reported by Kavyashree *et al.* (2023), Abha *et al.* (2024), Deeksha *et al.* (2024) and Niji *et al.* (2025).

The presence of high heritability and high genetic advance suggests that phenotypic selection for increased number of seeds per pod would be effective. This trait is an important component of seed production, and genotypes exhibiting higher values can be considered promising for selection and utilization in breeding programmes aimed at enhancing seed yield.

4.1.2.12. Grain yield per ha (kg/ha)

Grain yield per hectare exhibited a range from 397.04 to 1891.2 kg/ha, with an average yield of 841.41 kg/ha. The trait exhibited high GCV (33.09%) and PCV (40.04%) reflecting substantial variability among the genotypes with a relatively wide difference between the two estimates, indicating considerable environmental influence on the expression of grain yield. The wider difference between GCV and PCV indicates environmental influence, high heritability (68.29%) and high GAM (56.41%) suggests that additive gene action is important, making selection effective for improving grain yield. Similar findings were reported by Kavyashree *et al.* (2023), Poornima *et al.* (2023) and Deeksha *et al.* (2024).

The combination of high heritability and high GAM suggests that considerable improvement in grain yield can be achieved through selection, although environmental influence should be considered while evaluating genotypic performance. The wide range observed for grain yield further indicates the presence of diverse genetic material and provides an opportunity to identify superior genotypes with high yield potential.

4.1.3. Principal component analysis for seed yield and its attributing traits.

Principal component analysis (PCA) was carried out to assess the genetic diversity among the 107 cowpea genotypes based on twelve morphological, phenological, seed and yield related traits. The analysis identified five principal components with eigenvalues greater than one, together explaining 69.05 per cent of the total variation observed among the genotypes (Table 6 and Fig.4). Among these, PC1 contributed 20.26 per cent, followed by PC2 (16.67%), PC3 (12.74%), PC4 (9.78%) and PC5 (9.58%). These five components together explained a major proportion of the total variability, indicating that they effectively summarized the variation present in the experimental material. The contribution of the first component was relatively higher than that of the other components, indicating that the traits associated with PC1 were the major sources of variation among the genotypes. The cumulative contribution of the five components also indicates that a substantial proportion of the total variation could be explained by a relatively small number of independent components.

The contribution of individual traits to each principal component revealed that PC1 was mainly associated with number of pods per plant (NPPP), number of branches (NB), grain yield per hectare (GYPH) and number of seeds per pod (NSPP) (Table 7 and Fig.5 and Fig. 6). Hence, the first principal component largely represented yield and its associated characters. Thus, PC1 was mainly associated with yield and its component characters. The occurrence of several yield related traits together in the first component suggests that these characters contributed considerably to the overall differentiation of the genotypes. This variation could be useful in identifying genotypes with desirable combinations of yield-contributing characters. PC2 was influenced by seed breadth (SB), seed length (SL) and seed thickness (ST). The predominance of these seed related traits in PC2 indicates that seed size and dimensions formed an important source of variation independent of the major yield-related variation represented by PC1. This is of particular importance in the present study because considerable variation was also observed for seed characters in the germplasm material.

PC3 was mainly associated with days to 50 per cent flowering (DFF) and number of seeds per pod (NSPP), indicating that flowering behaviour and seed-bearing ability contributed to another dimension of variation. PC4 was largely influenced by days to maturity (DM), seed weight (SW) and pod length (PL), while PC5 received relatively higher contributions from plant height (PH), pod length (PL), seed weight (SW) and days to 50 per cent flowering (DFF). The distribution of different groups of characters across the components indicates that the genetic diversity in the present material was not controlled by a single group of traits but was contributed by phenological, morphological, seed and yield related characters.

The importance of yield related and seed traits in explaining genetic variation has also been reported in cowpea. Gerrano *et al.* (2011) and Mofokeng *et al.* (2020) observed considerable variation among cowpea genotypes through PCA, while similar importance of yield and related characters in explaining multivariate variation was reported by Gerrano *et al.* (2015) and Vijayakumar *et al.* (2020). The differences in the traits contributing to individual components are attributed to the genetic composition of the germplasm, number of genotypes evaluated and the set of characters included in the analysis.

The PCA scatter plot showed a wide distribution of the genotypes across different quadrants, further supporting the presence of considerable diversity in the material. Genotypes positioned close to each other are expected to have relatively similar performance for the traits included in the analysis, whereas genotypes located farther apart represent greater multivariate differences. The wide dispersion observed in the present study indicates that the germplasm contains diverse types that can be explored for selection and hybridization.

However, the PCA position of a genotype should be considered along with its actual trait performance rather than being used alone for selecting parents. Genotypes combining desirable agronomic performance with greater divergence from selected parents can provide useful material for future breeding programmes.

4.2 To study the genetic divergence and associationship between grain yield and its attributing quantitative traits.

Divergence among the 107 cowpea genotypes was evaluated using Mahalanobis D² statistics. The genotypes were grouped into different clusters and the average intra-cluster and

inter-cluster distances were estimated to assess the extent of genetic diversity among the clusters.

4.2.1. Clustering pattern of cowpea genotypes

4.2.2. Intra and inter-cluster distances

4.2.3. Phenotypic correlation coefficient for seed yield and its attributing traits

4.2.4. Path coefficient analysis for seed yield and yield attributing traits

4.2.1. Genetic diversity for seed yield and its attributing traits

Based on the Mahalanobis D² analysis, the 107 cowpea genotypes were classified into 5 clusters using generalized squared distances. The distribution of genotypes across the clusters is presented in Table 8 (Fig.7). Among the clusters, Cluster II included the maximum number of genotypes (48), followed by Cluster III (44 genotypes), Cluster V (10 genotypes), Cluster I (3 genotypes) and Cluster IV (2 genotypes). The clustering pattern revealed the existence of substantial genetic diversity among the evaluated cowpea genotypes. Similar clustering patterns indicating substantial genetic diversity among cowpea genotypes were also reported by Praveena *et al.* (2019), Walle *et al.* (2019), Purohit *et al.* (2020) and Oo *et al.* (2020), indicating the presence of substantial genetic diversity within cowpea germplasm. The unequal distribution of genotypes among the clusters indicates that the genotypes differed considerably in their overall performance across the characters studied. The presence of small clusters, particularly Cluster I and Cluster IV, suggests that some genotypes were relatively distinct from the larger groups.

The grouping of a large proportion of genotypes into Clusters II and III indicates that many genotypes shared relatively similar combinations of the characters considered. At the same time, the occurrence of genotypes in smaller clusters suggests the presence of relatively distinct genetic types. Such genotypes can be particularly useful when combined with information on cluster means and inter-cluster distances for identifying parents for hybridization.

4.2.2. Intra and inter-cluster distances.

The average intra and inter-cluster distances estimated using Mahalanobis D² statistics are presented in Table 9 (Fig. 8). The intra-cluster distances were consistently lower than the inter-cluster distances, indicating greater similarity among genotypes within the same cluster and higher genetic divergence between genotypes belonging to different clusters. The intra-cluster distance ranged from 0.00 to 7.995 units, with Cluster I recording the highest intra-cluster distance (7.995 units), followed by Cluster II (4.652 units), Cluster III (3.732 units), Cluster IV (3.693 units) and Cluster V (0.00). The relatively higher intra-cluster distance in Cluster I indicates that even though the genotypes were grouped together, they still possessed some degree of variation for the traits included in the analysis.

Among the inter-cluster distances, the maximum distance was observed between Cluster I and Cluster IV (9.354), indicating the greatest divergence between these two groups. The minimum inter-cluster distance was observed between Cluster II and Cluster III (2.209), suggesting that these two clusters were relatively similar in their overall character combinations. The large divergence between Clusters I and IV indicates that genotypes from these groups could be considered as potential parents. However, divergence alone does not guarantee superior recombinants; therefore, the performance of the genotypes for individual desirable traits should also be considered before selecting parents.

The presence of considerable inter-cluster divergence in the present material is in agreement with the findings of Praveena *et al.* (2019), Walle *et al.* (2019), Purohit *et al.* (2020) and Oo *et al.* (2020), who also observed significant differences among cowpea clusters. The variation in the magnitude of distances among studies is due to differences in the germplasm used and the number and type of characters included in the analysis.

4.2.3. Cluster means for different characters

The cluster mean values for twelve quantitative characters revealed considerable variation among the five clusters presented in Table 10, indicating the presence of diversity for the traits under study. Days to 50 per cent flowering varied from 43 days in cluster III to 53 days in cluster V. Cluster III recorded the earliest flowering, whereas cluster V showed comparatively late flowering. Days to maturity ranged from 89 days in cluster III to 99 days in cluster I, indicating that the genotypes grouped under cluster III were comparatively early maturing. The presence of early flowering and early maturity in Cluster III may be useful for breeding programmes targeting short-duration cowpea varieties.

Plant height showed substantial variation among the clusters, with the highest mean value of 77.06 cm in cluster V and the lowest value of 57.86 cm in cluster IV. The number of branches ranged from 7 in cluster IV to 11 in cluster V. The higher branching ability of Cluster V contribute to greater reproductive site formation and explain its superior performance for yield-related traits. For seed length, cluster I recorded the highest mean value (8.66 mm), while cluster II exhibited the lowest value (6.22 mm). Similarly, seed breadth was maximum in cluster I (6.60 mm) and minimum in cluster II (4.76 mm). Seed thickness was highest in cluster I (6.27 mm), whereas clusters III, IV and V recorded the lowest and identical mean value of 3.99 mm. The higher seed dimensions of Cluster I indicate that this group contains relatively large-seeded genotypes, which is useful when large seed size is a desirable breeding objective.

Considerable differences were also noticed for seed weight, which ranged from 8.67 in cluster II to 12.82 in cluster V. Pod length was maximum in cluster II (16.17 cm), while cluster IV recorded the minimum value (13.90 cm). The relatively higher seed weight in Cluster V, together with its higher plant height, number of branches and number of seeds per pod, indicates a favourable combination of several yield associated characters in this group. The number of pods per plant ranged from 12 in cluster IV to 14 in cluster I. For number of seeds per pod, cluster V exhibited a distinctly higher mean value of 20, whereas the lowest mean of 14 was recorded in cluster II. These differences shows that the clusters differed not only in seed characteristics but also in reproductive attributes.

Marked variation was observed among the clusters for grain yield per hectare. Cluster V recorded the highest mean grain yield per hectare (1645.79 kg/ha), followed by cluster III (964.25 kg/ha), cluster II (795.01 kg/ha), cluster I (727.03 kg/ha) and cluster IV (660.48 kg/ha). Thus, cluster V was characterized by higher mean values for plant height, number of branches, seed weight, number of seeds per pod and grain yield per plant. The concentration of several favourable yield-related traits in this cluster suggests that the genotypes belonging to cluster V may constitute useful material for the improvement of grain yield and its associated characters. Similar findings were recorded by Vavilapalli *et al.* (2014), Walle *et al.* (2019) and Oo *et al.* (2023).

4.2.4. Relative contribution of characters towards genetic divergence

The relative contribution of different quantitative characters towards genetic divergence showed that the traits differed considerably in their contribution to the total

diversity presented in Table 11 (Fig.9). Among the twelve characters studied, seed thickness contributed the maximum towards genetic divergence, accounting for 15.28 per cent of the total variation. This was followed by number of seeds per pod (11.85%), grain yield per hectare (11.52%), days to maturity (11.06%) and number of branches (10.84%). The relatively high contribution of seed thickness indicates that this trait was an important source of variation among the genotypes. The contribution of number of seeds per pod and grain yield indicates the importance of yield associated traits in differentiating the germplasm. Days to maturity also contributed considerably, suggesting that maturity duration differed sufficiently among the genotypes to influence their overall genetic divergence. The contribution of number of branches indicates that variation in plant architecture played an important role in differentiating the genotypes.

Plant height accounted for 7.32 per cent of the total divergence, while seed breadth and seed length contributed 6.83 and 6.36 per cent, respectively. Moderate contributions were observed for number of pods per plant (5.39%), pod length (5.31%) and seed weight (4.83%). Days to 50 per cent flowering showed the lowest contribution of 3.60 per cent towards the total genetic divergence. The present findings are in agreement with earlier reports of Vavilapalli *et al.* (2014), Walle *et al.* (2019) and Oo *et al.* (2023).

The first five major contributing characters, namely seed thickness, number of seeds per pod, grain yield per plant, days to maturity and number of branches, together accounted for more than half of the total divergence observed in the material. The results therefore indicate that these characters played an important role in differentiating the genotypes included in the study. The information on character contribution, when considered together with cluster means and inter-cluster distances, can be used in identifying genetically diverse and agronomically desirable parents for use in future cowpea improvement programmes.

Cluster V showed superior performance for grain yield, seed weight and number of seeds per pod, whereas Cluster I showed higher seed dimensions. Therefore, genotypes possessing desirable performance for these traits and belonging to genetically divergent clusters may be considered for hybridization. Such an approach helps in combining desirable agronomic and seed characteristics while also maintaining sufficient genetic diversity in the breeding material.

4.2.5. Phenotypic correlation coefficient for seed yield and its attributing traits

The correlation coefficient determines the degree and direction of association of different component characters with seed yield, thereby identifying traits that may be useful for genetic improvement of yield. Path coefficient analysis further partitions the correlation coefficients into direct and indirect effects of different component traits on seed yield. The phenotypic correlation coefficient values among seed yield and its attributing traits in 107 cowpea genotypes are presented in Table 12 (Fig. 10).

4.2.5.1. Days to 50 per cent flowering

Days to 50 per cent flowering showed positive but non-significant associations with plant height (0.116), seed breadth (0.082), seed thickness (0.058), seed weight (0.162), pod length (0.067), number of seeds per pod (0.091) and grain yield per hectare (0.006). Negative and non-significant correlations were recorded with days to physiological maturity (-0.080), number of branches per plant (-0.031), seed length (-0.052) and number of pods per plant (-0.166).

The negative association with number of pods per plant suggests that relatively early flowering genotypes tended to produce more pods, although the association was not significant. Thus, flowering time alone may not be an effective criteria for direct selection for higher grain yield, and it would be more appropriate to consider it along with major yield contributing traits.

The present finding differs from Nagalakshmi *et al.* (2020), who reported a significant positive association of days to 50 per cent flowering with seed yield (0.19) and a positive direct effect (0.07). In contrast, Chaudhary *et al.* (2020) reported a negative direct effect of days to flowering on seed yield, which is in agreement with the direction of the direct effect observed in the present study. The differences among studies may be due to differences in genetic material and growing environments.

4.2.5.2. Days to physiological maturity

Days to physiological maturity showed positive but non-significant correlations with pod length (0.154), seed thickness (0.110), seed breadth (0.099) and number of pods per plant

(0.001). In contrast, non-significant negative associations were noticed with number of seeds per pod (-0.121), seed weight (-0.119), plant height (-0.055), number of branches per plant (-0.028), seed length (-0.028) and grain yield per hectare (-0.021).

The negative association with grain yield indicates that longer maturity duration did not provide an advantage for grain yield in the present material. Similar negative association of days to maturity with seed yield was reported by Waghmare *et al.* (2019), Chaudhary *et al.* (2020) and Nagalakshmi *et al.* (2020). Thus, the slight negative correlation with yield appears to be influenced more by relationships with other component traits than by the direct effect of maturity itself.

4.2.5.3. Plant height (cm)

A significant positive relationship was recorded between plant height and number of branches per plant (0.518). The correlations of plant height with pod length (0.088), seed thickness (0.076), grain yield per hectare (0.061), seed weight (0.061), number of pods per plant (0.049) and number of seeds per pod (0.018) were positive but non-significant. However, seed breadth (-0.088) and seed length (-0.051) exhibited non-significant negative associations with plant height.

The present result is in agreement with Chaudhary *et al.* (2020) and Nagalakshmi *et al.* (2020). In contrast, Waghmare *et al.* (2019) reported a negative association between plant height and seed yield, indicating that the relationship may vary depending on the genetic material and growing conditions. The positive relationship with branching suggests that taller plants tend to possess greater branching ability. The weak positive association with grain yield suggests that plant height had a favourable relationship with yield.

4.2.5.4. Number of branches per plant

For the trait, number of branches per plant significant correlations were observed with number of pods per plant (0.517), grain yield per hectare (0.417), number of seeds per pod (0.249) and pod length (0.248). Seed weight (0.110), seed length (0.080), seed thickness (0.033) and seed breadth (0.029) recorded positive coefficients and these relationships were non-

significant. Chaudhary *et al.* (2020), Kavyashree *et al.* (2023) and Abha *et al.* (2024) reported a positive correlation of seed yield with branches per plant.

The significant positive association with grain yield indicates that increased branching was associated with greater yield in the present genotypes. This may be due to the greater number of reproductive sites produced by well-branched plants, resulting in increased pod formation. The positive relationship of branches with pods per plant and seeds per pod further supports this explanation. The consistency of the positive relationship across studies indicates that branching ability can be considered an important yield associated trait in cowpea.

4.2.5.5. Seed length (mm)

Seed length was strongly and positively correlated with seed breadth (0.764). Significant positive associations were also recorded with seed thickness (0.305) and number of pods per plant (0.208). A weak positive and non-significant relationship was observed with pod length (0.076). In comparison, grain yield per hectare (-0.029), seed weight (-0.045) and number of seeds per pod (-0.013) displayed negative, non-significant correlations.

The strong positive association among seed length, seed breadth and seed thickness indicates that these seed dimensional characters were closely related. However, the weak negative association with grain yield indicates that larger seed length did not necessarily result in higher total grain yield. Therefore, seed length may be more relevant for seed size preference and characterization than for direct selection for grain yield. Similar findings were reported by Carvalho *et al.* (2017), Porbeni *et al.* (2018) and Xiong *et al.* (2024)

4.2.5.6. Seed breadth (mm)

With respect to seed breadth, a significant positive association was evident with seed thickness (0.350). Positive but non-significant coefficients were obtained for number of pods per plant (0.082), grain yield per hectare (0.067), pod length (0.061) and number of seeds per pod (0.013). Seed weight was the only character showing a negative association (-0.032) which was also non-significant. Similar relationships were observed by Carvalho *et al.* (2017), Joshi *et al.* (2024), Xiong *et al.* (2024) and Mau *et al.* (2025).

The positive association between seed breadth and seed thickness indicates a close relationship between seed dimensions. Although the association of seed breadth with grain yield was positive, its non-significant nature suggests that seed breadth alone had limited association with yield.

4.2.5.7. Seed thickness (mm)

Pod length (0.069) and seed weight (0.010) were positively associated with seed thickness, whereas grain yield per hectare (-0.119), number of seeds per pod (-0.094) and number of pods per plant (-0.021) showed negative associations. The negative association of seed thickness with grain yield indicates that increased seed thickness was not accompanied by an increase in total grain yield. Similar relationships were observed by Joshi *et al.* (2024), Xiong *et al.* (2024) and Mau *et al.* (2025). Hence, seed thickness may be more useful as a seed characterization trait than as a direct criterion for grain yield improvement.

4.2.5.8. Seed weight (g)

For seed weight, all the subsequent associations were positive and non-significant. Similar coefficients were obtained with number of pods per plant (0.137) and number of seeds per pod (0.137), followed by grain yield per hectare (0.070) and pod length (0.022). The positive association with grain yield indicates that heavier seeds tended to be associated with higher yield, although the relationship was weak and non-significant. A similar positive association between seed weight and grain yield was reported by Manggoel *et al.* (2012), Nagalakshmi *et al.* (2020) and Kavyashree *et al.* (2023). Therefore, seed weight should be considered along with major yield components rather than used independently for selection.

4.2.5.9. Pod length (cm)

A significant positive correlation was found between pod length and number of pods per plant (0.243). Its relationships with grain yield per hectare (0.173) and number of seeds per

pod (0.075) were also positive; however, neither coefficient was statistically significant. Manggoel *et al.* (2012), Chaudhary *et al.* (2020) and Kavyashree *et al.* (2023) reported positive associations of pod length with yield. The positive relationship between pod length and number of pods per plant suggests that genotypes with longer pods produce more pods. The positive association with grain yield indicates a favourable trend, though the non-significant coefficient suggests that pod length alone is not sufficient to explain variation in yield.

4.2.5.10. Number of pods per plant

Among the characters considered, grain yield per hectare had a highly significant positive correlation with number of pods per plant (0.299). Number of seeds per pod also showed a positive association (0.100) although the relationship was non-significant. The significant positive association between pod number and grain yield indicates that pod production is an important component contributing to yield variation among the genotypes.

An increase in the number of pods provides a greater number of reproductive units and therefore has the potential to increase total grain production. This is supported by Manggoel *et al.* (2012), Aliyu *et al.* (2016), Chaudhary *et al.* (2020), and Kavyashree *et al.* (2023), who reported positive associations between pods per plant and yield. Thus, number of pods per plant can be considered an important trait for indirect selection for higher grain yield.

4.2.5.11. Number of seeds per pod

A strong and significant positive correlation (0.659) was recorded between number of seeds per pod and grain yield per hectare. Thus, number of pods per plant can be considered an important trait for indirect selection for higher grain yield. This is supported by Panchta *et al.* (2020), Chaudhary *et al.* (2020), and Kavyashree *et al.* (2023), who reported positive associations between seeds per pod and yield. Therefore, number of seeds per pod appears to be one of the most important yield associated traits in the present cowpea material.

4.2.6. Path coefficient analysis for seed yield and yield attributing traits

Path coefficient analysis provides a detailed assessment of the relationships between seed yield and its associated characters by separating the observed associations into direct contributions and indirect influences through other traits. The estimates of phenotypic direct and indirect effects of seed yield attributing characters on seed yield among the 107 cowpea genotypes are presented in Table 13 (Fig. 11).

4.2.6.1. Days to 50 per cent flowering

A negative direct contribution of days to 50 per cent flowering (-0.03196) towards seed yield was evident at the phenotypic level. However, favourable indirect influences were observed through number of seeds per pod (0.05254), seed breadth (0.02010), seed length (0.01254) and pod length (0.00350). Number of pods per plant (-0.02258), seed weight (-0.00775), number of branches (-0.00753), plant height (-0.00733), seed thickness (-0.00469) and days to maturity (-0.00109) contributed negatively through indirect pathways.

The negative direct effect indicates that an increase in flowering duration had an unfavourable direct influence on grain yield. However, the positive indirect effect through number of seeds per pod suggests that the overall association of flowering time with yield can be modified by its relationship with important yield components. Similar negative direct effect of days to 50 per cent flowering on seed yield was reported by Manggoel *et al.* (2012), Aliyu *et al.* (2016) and Chaudhary *et al.* (2020).

4.2.6.2. Days to physiological maturity

For days to maturity, the direct effect on seed yield was positive, though low in magnitude (0.01360). The indirect contribution through seed breadth (0.02444) was comparatively higher, followed by pod length (0.00802), seed length (0.00677), seed weight (0.00568), plant height (0.00348), days to 50 per cent flowering (0.00256) and number of pods per plant (0.00015). Number of seeds per pod (-0.07009), seed thickness (-0.00884) and number of branches (-0.00673) exerted negative influences.

The ² low magnitude of the direct effect indicates that maturity duration had only a limited direct contribution to grain yield. The relatively larger negative indirect effect through number of seeds per pod suggests that the association of maturity with yield is influenced considerably by its relationship with this important yield component. A similar positive direct effect of days to physiological ⁵⁷ maturity on seed yield was reported by Nagalakshmi *et al.* (2020), Manasa Lakshmi *et al.* (2021) and Kavyashree *et al.* (2023).

4.2.6.3. ¹⁸ Plant height (cm)

Plant height exerted a negative direct effect (-0.06319), its ³⁶ association with seed yield was positively influenced mainly through number of branches (0.12550). Positive indirect effects were noticed through seed length (0.01226), ¹¹ number of seeds per pod (0.01023), ² number of pods per plant (0.00672) and pod length (0.00457).

In contrast, indirect effects via seed breadth (-0.02176), seed thickness (-0.00609), ⁷ days to 50 per cent flowering (-0.00371), seed weight (-0.00292) and days to maturity (-0.00075) were negative. ²⁹ Negative indirect effects were observed through seed breadth, seed thickness, ⁷ days to 50 per cent flowering, seed weight ¹⁶ and days to maturity. The negative direct effect indicates that plant height itself did not favour grain yield, while its positive association with yield was mainly mediated through branching and other yield components.

4.2.6.4. ³² Number of branches per plant

A relatively high positive direct effect (0.24225) ⁵² was recorded for number of branches. ¹⁶ This character was influenced seed yield favourably through number of seeds per pod (0.14418), ² number of pods per plant (0.07043), pod length (0.01288), seed breadth (0.00722) ⁴⁰ and days to 50 per cent flowering (0.00099). ⁶² The negative indirect effects through plant height (-0.03274), seed length (-0.01950), seed weight (-0.00524), seed thickness (-0.00265) and days to maturity (-0.00038) reduced its overall contribution. Similar positive direct effects ¹⁶ of number of branches per plant on seed yield were reported by Sharma *et al.* (2017) and Manasa Lakshmi *et al.* (2021).

The ³⁵positive direct effect indicates that branching ability itself contributed favourably to grain yield. The additional positive effects through seeds per pod and pods per plant suggest that increased branching improve yield by increasing reproductive sites. ⁶Hence, number of branches per plant is considered an important selection criterion for yield improvement.

4.2.6.5. Seed length (mm)

Seed length was characterized by a comparatively large negative direct effect (-0.24285) on seed yield. Despite this unfavourable direct influence, ³⁴a positive indirect effect was expressed through seed breadth (0.18781). The pathways involving ¹²number of pods per plant (0.02827), ¹number of branches (0.01945), pod length (0.00396), plant height (0.00319), seed weight (0.00215) and days to 50 per cent flowering (0.00165) ³⁴were also positive. Negative indirect effects were contributed through seed thickness (-0.02440), ¹number of seeds per pod (-0.00748) and days to maturity (-0.00038). ⁶Negative indirect effects were observed through seed thickness, ¹number of seeds per pod and days to maturity. Similar findings were reported by Carvalho *et al.* (2017), Porbeni *et al.* (2018) and Xiong *et al.* (2024).

⁵⁹The negative direct effect indicates that greater seed length was not favourable for grain yield in the present material. ⁴⁸The positive indirect effect through seed breadth explains why the overall relationship between seed length and some other traits may not directly reflect its own contribution to yield. Therefore, selection for increased seed length alone can not be effective for improving grain yield.

4.2.6.6. Seed breadth (mm)

Among the seed dimensional characters, seed breadth produced ³⁴a high positive direct effect (0.24598) ¹on seed yield. Its favourable indirect influence was also expressed through number of pods per plant (0.01119), number of seeds per pod (0.00725), number of branches (0.00711), plant height (0.00559), pod length (0.00319), seed weight (0.00155) and days to maturity (0.00135). However, the indirect pathways through seed length (-0.18542), seed thickness (-0.02807) and days to 50 per cent flowering (-0.00261) contributed negatively. ⁶Similar findings were reported by Porbeni *et al.* (2018) and Xiong ³⁵*et al.* (2024).

The positive direct effect indicates that seed breadth had a favourable contribution to grain yield in the present material. Therefore, seed breadth have potential as a selection criterion, but its simultaneous relationship with other seed traits should be considered.

4.2.6.7. Seed thickness (mm)

The direct influence of seed thickness on seed yield was negative (-0.08011). This adverse effect was partly compensated by positive indirect contributions through seed breadth (0.08618), number of branches (0.00800), pod length (0.00360), days to maturity (0.00150) and number of pods per plant (0.00165). Conversely, negative effects were mediated through seed length (-0.07398), number of seeds per pod (-0.05449), plant height (-0.00481), number of pods per plant (-0.00280), days to 50 per cent flowering (-0.00187) and seed weight (-0.00046).

The negative direct effect indicates that seed thickness did not contribute favourably to grain yield when its own influence was separated from the effects of other traits. The positive indirect effect through seed breadth suggests that part of its association with yield arise from its relationship with other seed dimensions. This was in agreement in findings of Carvalho *et al.* (2017), Joshi *et al.* (2024), Xiong *et al.* (2024) and Mau *et al.* (2025).

4.2.6.8. Seed weight (g)

With respect to seed weight, the direct path towards seed yield was negative (-0.04778). The strongest favourable indirect contribution was observed through number of seeds per pod (0.07964), followed by number of branches (0.02654), number of pods per plant (0.01860), seed length (0.01092) and pod length (0.00115). Unfavourable indirect influences were transmitted through seed breadth (-0.00797), days to 50 per cent flowering (-0.00518), plant height (-0.00387), days to maturity (-0.00162) and seed thickness (-0.00078).

The negative direct effect indicates that seed weight did not make a favourable direct contribution to grain yield in the present material. A similar association was reported by Carvalho *et al.* (2017), Nagalakshmi *et al.* (2020) and Kavyashree *et al.* (2023). The positive phenotypic association observed earlier therefore be attributed mainly to its favourable

relationships with other yield components, particularly number of seeds per pod. This indicates that selection for seed weight alone may not necessarily increase grain yield.

4.2.6.9. Pod length (cm)

A positive direct effect of 0.05192 was obtained for pod length. Its contribution to seed yield was further enhanced by favourable indirect effects through number of branches (0.06009), number of seeds per pod (0.04353), number of pods per plant (0.03310), seed breadth (0.01512) and days to maturity (0.00210). However, negative indirect influences were evident through seed length (-0.01854), plant height (-0.00556), seed thickness (-0.00555), days to 50 per cent flowering (-0.00216) and seed weight (-0.00106).

Similar positive direct effect of pod length on seed yield was reported by Manggoel *et al.* (2012), Aliyu *et al.* (2016), Chaudhary *et al.* (2020), Kavyashree *et al.* (2023). The positive direct effect indicates that pod length had a favourable contribution to grain yield, although its magnitude was relatively low. The positive indirect effects through branches, seeds per pod and pods per plant further strengthened its contribution.

4.2.6.10. Number of pods per plant

Number of pods per plant contributed positively to seed yield through a direct effect of 0.13620. A nearly comparable indirect effect was mediated through number of branches (0.12527), while number of seeds per pod (0.05803), seed breadth (0.02022), pod length (0.01262), days to 50 per cent flowering (0.00530), seed thickness (0.00165) and days to maturity (0.00001) also provided positive contributions.

Negative indirect effects were confined to seed length (-0.05041), seed weight (-0.00652) and plant height (-0.00312). The positive direct effect indicates that pod number itself contributed favourably to grain yield. The substantial positive indirect effect through branches further suggests that improved branching increase yield partly through increased pod production. A similar positive direct effect of number of pods per plant on seed yield was reported by Sharma *et al.* (2017), Nagalakshmi *et al.* (2020), Manasa Lakshmi *et al.* (2021) and Kavyashree *et al.* (2023).

4.2.6.11. Number of seeds per pod

Of all the component characters examined, number of seeds per pod exhibited the highest positive direct effect (0.57965) on seed yield. Its favourable influence was supplemented by positive indirect effects through number of branches (0.06026), number of pods per plant (0.01364), seed thickness (0.00753), pod length (0.00390), seed length (0.00313) and seed breadth (0.00308).

In comparison, negative indirect contributions through seed weight (-0.00657), days to 50 per cent flowering (-0.00290), days to maturity (-0.00164) and plant height (-0.00112) were relatively small. The magnitude of the direct effect indicates that number of seeds per pod was the most important character contributing directly to grain yield in the present material. This result is particularly important because the trait also recorded the highest positive phenotypic correlation with grain yield. The agreement between the correlation and path analysis provides strong evidence for the importance of this trait in selection. A similar direct effect of number of seeds per pod on seed yield was reported by Panchta *et al.* (2020) and Kavyashree *et al.* (2023).

The residual effect of 0.66 indicated that the characters included in the path analysis collectively explained approximately 56.44 per cent of the variation in grain yield, while the remaining 43.56 per cent was attributable to other factors not included in the analysis and environmental or random effects. Overall, the path analysis indicated that number of seeds per pod, number of branches per plant, seed breadth and number of pods per plant exerted positive direct effects on grain yield, whereas days to 50 per cent flowering, plant height, seed length, seed thickness and seed weight showed negative direct effects. Among all the traits, number of seeds per pod had the highest positive direct effect, indicating its major role in determining grain yield. The positive direct effects of number of branches and number of pods per plant further emphasized the importance of reproductive and branching traits in yield improvement. Therefore, these traits particularly number of seeds per pod, can be given greater emphasis while selecting superior genotypes for grain yield improvement.

4.3. To study the seed morphometric and quality traits of cowpea genotypes.

The seed morphological and quality traits of the 107 cowpea genotypes were evaluated by recording seed size, seed shape, seed color presented in Table 14. Among 107 genotypes, based on observations taken, only the top ten best genotypes, along with checks, are evaluated for germination percentage. Seedling vigour was evaluated by calculating the Seedling Vigour Index using germination percentage and seedling length presented in Table 15. Considerable variation in seed morphological traits, including seed size, shape and colour, among cowpea genotypes has also been reported by Adewale *et al.* (2010), Carvalho *et al.* (2017), Porbeni *et al.* (2017), Santos *et al.* (2018) and Zafeiriou *et al.* (2023).

4.3.1. Seed shape, size and color of cowpea genotypes

4.3.2. Germination percentage and seedling vigour of ten selected cowpea genotypes

4.3.2. Germination percentage and seedling vigour of ten selected cowpea genotypes

Seed quality and early seedling establishment are important attributes influencing the successful establishment and subsequent performance of crop plants. In the present investigation, 10 best cowpea genotypes along with checks were evaluated for germination percentage and seedling vigour.

Considerable variation was observed among the cowpea genotypes for germination percentage, seedling vigour and grain yield (Table 15). The germination percentage ranged from **87.50 to 97.30 per cent**, seedling vigour ranged from **2350.30 to 3054.03**, and grain yield ranged from **820.12 to 1891.22 kg/ha**. The variation observed for these characters indicates differences among the genotypes with respect to seed quality, early seedling establishment and yield potential. Such variation provides an opportunity for identifying genotypes possessing desirable seed quality along with superior field performance.

Among the genotypes evaluated, **V-16 recorded the highest grain yield (1891.22 kg/ha)** along with **97.30 per cent germination** and a seedling vigour index of **2732.24**. **EC-458480** was the next best genotype for grain yield, recording **1876.32 kg/ha**, with a high germination percentage of **96.80 per cent** and the **highest seedling vigour (3027.84)** among the genotypes evaluated. The relatively superior performance of these genotypes for both seed

quality and grain yield indicates their potential as promising genetic material for further evaluation and utilization in cowpea improvement programmes.

IC-2591054 and EC-458418 also recorded relatively high grain yields of **1640.44 kg/ha and 1633.63 kg/ha** respectively, with germination percentages of 89.50 per cent and 91.80 per cent. **C4-IT-38956-1** recorded a grain yield of 1538.67 kg/ha along with high seedling vigour (2996.39) while **NBC-30 and IC-402172** recorded grain yields of 1446.52 and 1438.52 kg/ha respectively. These genotypes can be considered as promising material based on their comparatively favourable performance for the traits studied.

Among the check genotypes, **KBC-2 recorded the highest seedling vigour (3054.03)**, whereas KBC-9 recorded a comparatively high germination percentage (94.30%). However, their grain yields were 904.52 and 983.33 kg/ha respectively. The local variety and C-152 also recorded relatively high germination percentages of 90.00 and 94.30 per cent, respectively, but their grain yields were lower, at 873.85 and 820.12 kg/ha. This indicates that high germination percentage or seedling vigour alone does not necessarily result in high grain yield, as final yield is influenced by several growth, reproductive and yield contributing traits. Similarly, variability in germination percentage and seedling vigour among cowpea genotypes was reported by Gerrano *et al.* (2017) and Kamara *et al.* (2019).

V. SUMMARY AND CONCLUSION

The present investigation¹ was carried out to assess the extent of genetic variability and diversity available in cowpea (*Vigna unguiculata* L. Walp.) germplasm and to understand⁹ the association of different component traits with grain yield. The study also aimed to identify important yield-contributing characters through correlation and path coefficient analyses and to characterize the genotypes for seed morphometric and quality traits. A total of 107 cowpea genotypes comprising 102 germplasm lines and five checks, namely C-152, KBC-2, KBC-9, IT-38956-1 and a local variety, were evaluated during Kharif 2025 in an Augmented Randomized Block Design with six blocks. Observations recorded on twelve quantitative traits¹ were subjected to analysis of variance, genetic variability analysis, principal component analysis, Mahalanobis D² analysis,⁶ correlation and path coefficient analysis. Seed colour, shape and size were also recorded, while selected promising genotypes along with checks were further assessed for germination and seedling vigour.

The analysis of variance indicated¹³ significant differences among the genotypes for the characters studied, confirming the presence of adequate variability in the experimental material. The magnitude of variability differed among the traits, as reflected by the⁸ estimates

of GCV, PCV, heritability⁶³ and genetic advance as per cent of mean. In general, PCV estimates¹³ were slightly higher than the corresponding GCV estimates. The difference between the two estimates was narrow for most characters, suggesting that the observed variation was largely due to genetic differences among the genotypes, with relatively less⁹ environmental influence on the expression of these traits.

Plant height recorded high GCV and PCV (21.08% and 21.12%) together with high heritability (99.64%)⁶³ and high genetic advance as per cent of mean (43.41%), indicating good scope for improvement through selection, while¹ number of branches per plant recorded GCV (27.19%), PCV(27.33%), heritability (99.03%) and GAM(55.83%). Grain yield per hectare also showed a high magnitude of variation (GCV 33.09%; PCV 40.04%) with high heritability (68.29%) and high GAM (56.41%), indicating considerable scope for genetic improvement.

²⁰ Days to 50 per cent flowering, seed thickness,⁹ seed weight, pod length, number of pods¹³ per plant and number of seeds per pod recorded moderate GCV along with high heritability and⁶ moderate to high GAM (GCV 10.78–17.88%; heritability 80.88–96.47%; GAM 20.29–34.84%), suggesting favourable prospects for improvement through selection. In contrast, days to physiological maturity showed relatively low variability (GCV 5.16%; PCV 5.21%; GAM 10.78%), indicating limited variability for this character.

Principal component analysis provided further insight into the pattern of variation present among the genotypes. Five principal components with eigenvalues greater than one together accounted for 69.05 per cent of the total variation.⁸ The first principal component explained 20.26 per cent of the variation and was mainly influenced by⁶⁵ number of pods per plant (28.86%),¹ number of branches per plant (27.99%), grain yield per hectare (15.53%)¹² and number of seeds per pod (25.58%). Seed dimensional traits, particularly seed breadth (30.45%), seed length (29.45%) and seed thickness (19.42%), contributed predominantly to the second principal component.¹ The distribution of genotypes in the PCA plot also showed considerable dispersion, reflecting the diverse nature of the experimental material.

The Mahalanobis D² analysis grouped the 107 genotypes into five⁷⁴ clusters. Cluster II was the largest with 48 genotypes, closely followed by Cluster III with 44 genotypes, whereas Cluster V,⁹ cluster I and Cluster IV consisted of ten, three and two genotypes, respectively. Inter-cluster distances were generally greater than intra-cluster distances, indicating greater

divergence between genotypes placed in different clusters. The intra-cluster distance ranged from 0.00 to 7.995 units, with the maximum intra-cluster distance recorded in Cluster I (7.995 units). The widest genetic distance was observed between Cluster I and Cluster IV (9.354 units), suggesting that genotypes belonging to these two clusters represent comparatively diverse genetic backgrounds. Such divergent genotypes are useful in future crossing programmes for creating wider variability and recovering desirable recombinants. The lowest inter-cluster distance was observed between Cluster II and Cluster III (2.209 units), showing greater similarity between these two groups.

Cluster means revealed substantial differences among the five clusters for the characters studied. Cluster V recorded the highest mean values for plant height (77.06 cm), number of branches per plant (11.32), seed weight (12.82 g), number of seeds per pod (20.20) and grain yield per hectare (1645.79 kg/ha). Cluster III recorded the earliest flowering (43.16 days) and earliest maturity (88.62 days) and also had relatively high grain yield (964.25 kg/ha). Cluster I recorded the highest seed length (8.66 mm), seed breadth (6.60 mm) and seed thickness (6.27 mm), whereas Cluster II recorded the maximum pod length (16.17 cm). Thus, Cluster V was particularly promising for yield and its associated traits, whereas Cluster I was characterized by comparatively larger seed dimensions.

The relative contribution of individual characters towards genetic divergence showed that seed thickness contributed the maximum (15.28%), followed by number of seeds per pod (11.85%), grain yield per hectare (11.52%), days to physiological maturity (11.06%) and number of branches per plant (10.84%). These five characters together contributed 60.55 per cent of the total divergence. Plant height, seed breadth and seed length contributed 7.32, 6.83 and 6.36 per cent, respectively, while days to 50 per cent flowering contributed the least (3.60%). These results indicate that seed thickness, number of seeds per pod, grain yield, maturity duration and number of branches were important in differentiating the genotypes.

The correlation analysis showed that grain yield per hectare was positively and significantly associated with number of seeds per pod (0.659*), number of pods per plant (0.299) and number of branches per plant (0.417). Among these characters, number of seeds per pod exhibited a particularly strong positive relationship with grain yield. Pod length (0.070), seed weight (0.070), seed breadth (0.067) and plant height (0.061) also showed positive associations with yield, although their correlations were not significant. On the other hand, days to physiological maturity (-0.021), seed length (-0.029) and seed thickness (-0.119)

showed weak negative associations with grain yield. The results suggest that improvement in number of seeds per pod, number of pods per plant and number of branches per plant can contribute favourably towards achieving higher grain yield.

The path coefficient analysis clarified the contribution of individual characters towards grain yield. Number of seeds per pod recorded the highest positive direct effect on yield (0.57965). Positive direct effects were also observed for seed breadth (0.24598), number of branches per plant (0.24225), number of pods per plant (0.13620), pod length (0.05192) and days to physiological maturity (0.01360). The strong positive association of number of seeds per pod with grain yield, together with its high positive direct effect, highlights its importance as a useful character for yield improvement. Number of branches per plant and number of pods per plant also emerged as important yield components. Selection based on these characters therefore be useful in identifying promising high yielding cowpea genotypes.

Considerable variation was also noticed for seed morphological characters. The genotypes differed widely in seed colour, with cream, brown, black, reddish brown, purplish brown, purple, white and speckled seed types being observed. Variation was also present for seed shape and size. Most of the genotypes possessed rhomboid seeds, while a few were characterized by kidney shaped seeds. Based on seed size, the genotypes were categorized into small, medium and large or bold seeded types. This variation in visible seed characters is useful for germplasm characterization and may also help in selecting genotypes according to specific breeding or consumer preferences.

The evaluation of seed quality traits among selected promising genotypes and checks also revealed noticeable differences. Germination percentage was highest in V-16 (97.3%), whereas genotype EC-458480 recorded the highest seedling vigour index (3027.84), whereas EC-458480 recorded the highest seedling vigour (3027.84) along with high germination (96.80%) and grain yield (1876.32 kg/ha). Among the checks, KBC-2 recorded the highest seedling vigour (3054.03) but relatively lower grain yield (904.52 kg/ha), indicating that high seedling vigour alone may not necessarily result in high final grain yield. Differences in germination and seedling vigour among the evaluated entries indicated variation in seed quality, providing an additional basis for identifying promising genotypes.

The findings of the present investigation demonstrate the availability of useful variability and diversity within the evaluated cowpea germplasm. Number of seeds per pod

emerged as an important character because of its strong positive association and high direct contribution towards grain yield.

Number of branches per plant and number of pods per plant were also positively associated with yield and may be considered while selecting superior genotypes. The diversity observed through Mahalanobis D^2 analysis and PCA provides an opportunity to identify genetically diverse parents for future breeding programmes. The variation recorded for seed morphology, germination and seedling vigour further indicates the possibility of selecting genotypes combining desirable agronomic performance with good seed characteristics.

Future line of work

Based on the findings of the present investigation, the following future line of work is suggested:

1. Promising genotypes identified in the present study may be evaluated over different seasons and locations to confirm their performance and adaptability under varying environmental conditions.
2. Genotypes belonging to highly divergent clusters, particularly those showing desirable agronomic performance, may be utilized as parents in hybridization programmes for generating useful variability and obtaining superior recombinants.
3. Promising genotypes combining high yield with desirable seed morphology, germination and seedling vigour may be further evaluated for their potential use in cowpea improvement programmes.

